

## VA 3/1 hybrid ARM Rates

- 1) Populate the historical time series until asOf = 20180401 (applicationDate) as below:  
select l.firstPaymtDt, factorDate = dateadd(mm,-1,l.firstPaymtDt), asOf = dateadd(mm,-4,l.firstPaymtDt), sum(l.originalLoanAmount), wac = sum(l.originalLoanAmount \* l.origNoteRate)/sum(l.originalLoanAmount), count(\*)  
from gnm.loan l, gnm.secs s  
where s.seclId = l.seclId  
and l.loanType in ('V')  
and s.prefix in ('AT','TL')---3/1  
and applicationDate <='20180401'  
group by l.firstPaymtDt, factorDate, asOf  
order by asOf DESC
- 2) The above query assumes implicitly a mortgage rate lag of two months.
- 3) **To construct the appropriate mortgage rate series assuming a six week lag, we compare this series with the six week lag series derived from the Ginnie Mae 30 year mortgage rate, compute the two month average spread between the two and then apply the new spread to the Ginnie Mae 30 year mortgage rate. The result is a mortgage rate series for the VA 3/1 hybrid ARM which is closer to a six week lag assumption.**
- 4) For asOf after 20180401, we assume a constant spread of **140** bps below the corresponding GNMA 30 yr fixed mortgage rate.
- 5) For mortgage spread model, we will assume an equilibrium spread of **-158** bps (**-140** bps vs LT Ginnie Mae mortgage rate) and lambda decay of 0.8909.
- 6) The mortgage rate model will then pass in a lagged mortgage rate (average of 1 month and 2 month rate) into the prepayment model